

RE07 Global Infrastructure Investment and Management

Week 3

Private Investment in Infrastructure

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Guest presentations: Moody's Analytics and S&P Global Ratings

What this week does

Week 2 described the institutions that decide what gets built. This week asks who pays for it once the state has decided it will not, and why the answer is so much narrower than the investment need implies. The week has a specific empirical spine: what we actually know about the returns to infrastructure equity and the credit performance of infrastructure debt, and how much less we know about the first than the second.

A	The investor's problem	30 min
B	Routes into the asset	30 min
C	What the return evidence shows	40 min
D	Infrastructure debt: credit performance and rating	45 min
E	Why capital does not reach the need	35 min
F	Public support and blended finance	30 min
G	The financialisation critique	25 min

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE02 Lent lectures 5 to 8, private real estate vehicles and strategies. RE01 sessions 8 and 9, portfolios and risk factor models; sessions 11 and 12, term structure and bond ratings; sessions 15 and 16, REITs.	The four Lent lectures on private vehicles are the closest match to anything in this module. The listed and unlisted question, the fee load and the fund life all transfer directly.
Planning, Growth and Regeneration	PGR05 lecture 8a, financialisation, centralisation and regional inequality. PGR07 lecture 3, financing mechanisms behind regeneration. PGR02 Lent week 6, the financialisation of housing.	The critical account of what happens when an essential service becomes a financial asset. Block G is built on it, and the housing parallel is close enough to be uncomfortable.
Environmental Policy	EP06 lectures 5 and 6, technology innovation economics, knowledge spillovers and the valley of death. EP01 Lent week 4, carbon markets and climate finance. EP02, environmental regulation and its impact on innovation.	The valley of death is the first-of-a-kind financing gap in another vocabulary, and climate finance is the fastest growing source of the capital this week is about.
Everyone	RM01 lecture 6, qualitative dependent variable models.	A default is a binary outcome. Every credit model in block D is a qualitative dependent variable model with a great deal of institutional detail attached.

Block A

The investor's problem

Who has the money, what they owe, and why the second question determines the first.

Indicative length 30 minutes

Start from the liability, not from the asset

Infrastructure is described in fund marketing as an attractive asset class. That is the wrong end of the problem. An institution allocates capital because of what it owes, not because of what an asset is like, and the same asset is appropriate for one investor and inappropriate for another with no change in its characteristics.

A mature pension scheme paying inflation-linked pensions for forty years has a liability that a regulated network matches unusually well. An insurer writing annuities has a similar problem and a different capital regime. An endowment with no fixed obligations has no such need and should buy infrastructure only if it expects to be paid for the illiquidity. Read every allocation back to the liability and the apparent inconsistencies in market behaviour resolve.

Who supplies the capital

Pension funds

Long-dated and often inflation-linked liabilities. The natural holder in theory, constrained in practice by scale, governance capacity and the cost of building expertise.

Insurers

Long liabilities and a capital regime that treats qualifying infrastructure debt more favourably than equity, which shapes what they buy far more than preference does.

Sovereign wealth funds

No liabilities in the ordinary sense, very long horizons, and the scale to invest directly and take control positions alongside operators.

Listed funds and corporates

Utilities, developers and conglomerates holding assets as strategic rather than portfolio positions, with permanent capital and no fund life.

Banks

Still the dominant provider of construction-phase debt. Appetite is set by capital regulation and by balance sheet capacity rather than by the asset.

Development institutions

Deploying where commercial capital will not, on a mandate that is not a required return. Block F is about what that does to a transaction.

What each investor actually needs

Investor	Horizon	What matters most	What rules them out of a transaction
Mature pension scheme	Twenty years plus	Inflation linkage and cash yield	Construction risk, and any structure with a forced exit
Life insurer	Fifteen to thirty years	Rated, long-dated, investment grade debt	Unrated exposures and equity capital charges
Sovereign wealth fund	Indefinite	Scale, control and direct access	Small transactions that cannot absorb the ticket size
Infrastructure fund	Ten years, with extensions	A credible exit within the fund life	Assets with no buyer at exit, and regulated returns too low for the fee load
Bank	Five to fifteen years	Security, covenants and syndication capacity	Tenor beyond what the capital regime makes economic

Capital regulation shapes the market more than preferences do

Two regimes do most of the work. Bank capital rules under Basel treat project finance as specialised lending, with its own slotting criteria and risk weights, and the treatment of long-dated exposures affects tenor directly. Insurance capital rules distinguish qualifying infrastructure investments meeting defined criteria and apply a lower capital charge to them than to comparable corporate exposures.

The consequence is that a change in a technical annex can move billions of allocation without any change in the underlying assets. Students who want to understand why the market looks as it does should read the capital rules before the fund marketing.

Why allocations are smaller than the rhetoric

Infrastructure is described as a natural fit for institutional capital, and institutional allocations to it remain modest as a share of total assets. The reasons are practical rather than philosophical.

A small allocation cannot justify the internal expertise the asset requires, so it is made through funds, which carry a fee load that eats much of the premium. Trustees are accountable for decisions they cannot easily explain, and an illiquid, hard-to-value asset is difficult to defend when it underperforms. Consultants recommend within a peer range, which anchors allocations to what everyone else does. And the assets that meet institutional criteria are competed for by everyone at once, which compresses the return that justified the allocation.

Governance capacity is the binding constraint for most investors

- An infrastructure investment requires diligence on engineering, regulation, contract, tax and jurisdiction, none of which a generalist team can do well.
- Ongoing governance is harder than the purchase: board seats on a consortium, capital calls, refinancing decisions, and a regulatory determination every five years.
- The expertise is expensive and mobile, and public sector-owned plans compete for it with private managers who pay more.
- This is why intermediation exists, and why the fee is not simply a transfer. Week 5 asks what it is actually worth.
- It is also why scale determines strategy in this asset class more sharply than in listed markets.

Block A discussion

Seminar questions

1. A defined benefit scheme with a twenty-year liability profile is offered a ten-year closed-end infrastructure fund. What is the mismatch costing it, and what alternatives does it have at its size?
2. If insurance capital rules favour rated infrastructure debt, what does that imply for the assets that get built and the ones that do not?
3. Which of the investors on the previous slide would you want as a counterparty if you were a government seeking a forty-year partner, and why?

Block B

Routes into the asset

Listed, unlisted, co-investment, direct and debt, and what each costs in control and in fees.

Indicative length 30 minutes

Five routes, and what they cost

Route	Control	Fees	Liquidity	Minimum scale	Typical holder
Listed equity	None	Low	Daily	Very low	Any investor
Unlisted closed fund	None beyond fund governance	Management fee and carried interest	None until exit	Moderate	Mid-sized schemes and insurers
Co-investment	Limited	Reduced or none	None	High	Large pension plans
Direct	Full, or shared in consortium	Internal cost only	Negotiated sale	Very high	Sovereign funds, large plans, corporates
Debt	None, but covenants and step-in	Low	Limited secondary market	Moderate	Insurers, debt funds, banks

The fund model, stated precisely

A general partner raises commitments from limited partners, draws them over an investment period, buys and manages assets, and returns capital through exits within a fixed life, usually ten years with provision for extension. The general partner charges a management fee on committed or invested capital and takes a share of profits above a hurdle rate.

The model suits assets bought to be improved and sold. It suits assets bought to be held for forty years very badly, and that mismatch is the central empirical finding of this week. The fee load is the visible cost. The forced exit is the expensive one, because it converts a long-duration asset into a ten-year bet on the state of the market at the moment the fund must sell.

Listed and unlisted, compared honestly

Listed

- Observable prices and genuine liquidity
- Low minimum investment, so available to any investor
- Correlation with equity markets is visible, and rises in stress
- Exposure includes company-level leverage, management and unrelated businesses
- Reported volatility is real volatility

Unlisted

- Appraisal-based valuations, typically quarterly
- Asset-level exposure with control rights and information
- Reported volatility is smoothed by the appraisal process and understates the truth
- Reported correlation with equities is partly an artefact of the valuation method
- Fee load and fund life are the price of access

Direct investment, and why it is only available at scale

The largest Canadian and Australian plans built internal teams and buy assets directly, alone or in consortium. The saving is not only the fee. Direct ownership removes the fund life, so the holding period can match the asset, which is the thing the fund model cannot deliver at any price.

The cost is organisational. A team able to underwrite and then govern a regulated utility is expensive, hard to recruit and harder to retain against private market employers. Below the scale at which such a team can be justified, the fund route dominates, and the interesting question for most investors is not whether to go direct but how much co-investment they can obtain alongside a manager.

The strategy labels, and what they mean

Label	What is bought	Return sought	Risk actually taken
Core	Operating, regulated or contracted assets in stable jurisdictions	Modest, mostly cash yield	Regulatory and refinancing
Core plus	Operating assets with some volume exposure or growth capital need	Moderate, yield and growth	Demand and capital expenditure
Value add	Assets needing operational improvement, expansion or repositioning	Higher, mostly capital growth	Execution, and exit pricing
Opportunistic	Greenfield, distressed, or assets in difficult jurisdictions	Equity-like	Construction, political, and everything else
Debt	Senior or subordinated loans and private placements	Spread over a benchmark	Credit, and illiquidity

The exit problem, and the secondary market that is not quite there

A closed-end fund must sell. The buyer is another fund, a direct investor, or a listed vehicle, and in a weak market may be none of them. Unlike listed equity there is no continuous price, and unlike real estate there is no deep transaction record to price against.

A secondary market in fund interests has developed and is now substantial, which allows a limited partner to exit a position before the fund does. It does not solve the underlying problem, because the secondary buyer is buying the same forced-exit structure at a discount. Continuation vehicles, in which a manager sells an asset to a fund it also manages, address the mismatch and introduce an obvious conflict.

Why infrastructure debt funds grew so quickly

Two regulatory changes did most of the work. Bank capital rules made long-dated project lending more expensive to hold, which reduced bank appetite for the tenors infrastructure needs. Insurance capital rules created a favourable treatment for qualifying infrastructure debt, which created demand from investors who could not easily originate it themselves.

A debt fund sits in that gap: it originates or acquires loans, and sells the exposure to institutions in a form their capital regime rewards. Whether it adds value beyond arbitraging the two regimes is exactly the sort of question week 5 asks about intermediaries.

Block B discussion

Seminar questions

1. The same asset is bought listed, through a fund, and directly. Rank the returns the three investors receive and explain what accounts for the differences.
2. Co-investment offers reduced fees alongside a manager. Why would a manager offer it, and what does the answer tell you about which assets get offered?
3. If unlisted valuations are appraisal-based, in which direction is reported correlation with equities biased, and what does that do to a strategic asset allocation?

Block C

What the return evidence shows

The claims, the tests, and the uncomfortable distance between them.

Indicative length 40 minutes

The five claims, restated

- Stable cash flows, because demand for an essential service is inelastic and less cyclical than corporate earnings.
- Inflation linkage, because tariffs are indexed by regulation or by contract.
- Low correlation with equities, because returns are driven by regulatory and contractual factors rather than by market sentiment.
- Long duration, matching the liabilities of pension funds and insurers.
- An illiquidity premium, paid to investors who accept that they cannot sell.
- Each of these is a claim about a contract or a regulatory settlement rather than about the asset. That is where they should be tested, and it is also where they expire.

The vehicle, not the asset

10 years

Contractual life of the closed-end private funds through which institutions predominantly hold infrastructure.

Andonov, Kräussl and Rauh document what institutional investors say they want from infrastructure, which is long-term stable cash flow, and then what they actually buy, which is exposure through finite-horizon closed private funds.

The cash flows delivered by those funds display volatility and cyclicalities similar to other private equity, and performance depends on quick deal exits in much the same way.

The asset may last forty years. The vehicle lasts ten. Investors are exposed to the vehicle, and any claim about the asset that does not survive the wrapper is not a claim about their return.

Tail risk that the recent record conceals

1927

Start of the reconstructed United States listed infrastructure return series used to test the low-risk claim.

Bianchi, Bornholt, Drew and Howard reconstruct listed infrastructure returns back to 1927 and analyse them in a four-factor framework. Low market beta and the value premium explain most of the return.

Their finding on risk is what matters for allocation. Returns measured over recent decades understate long-run tail risk, which turns out to be comparable to that of a portfolio of equities. Some indices they examine are undesirable in a portfolio selection framework; others, notably broad and small utilities, do show diversification benefit.

The methodological lesson generalises to every asset class whose track record begins after 1990: it has not been observed through a full distribution of macroeconomic states.

Why the equity evidence is so much thinner than it should be

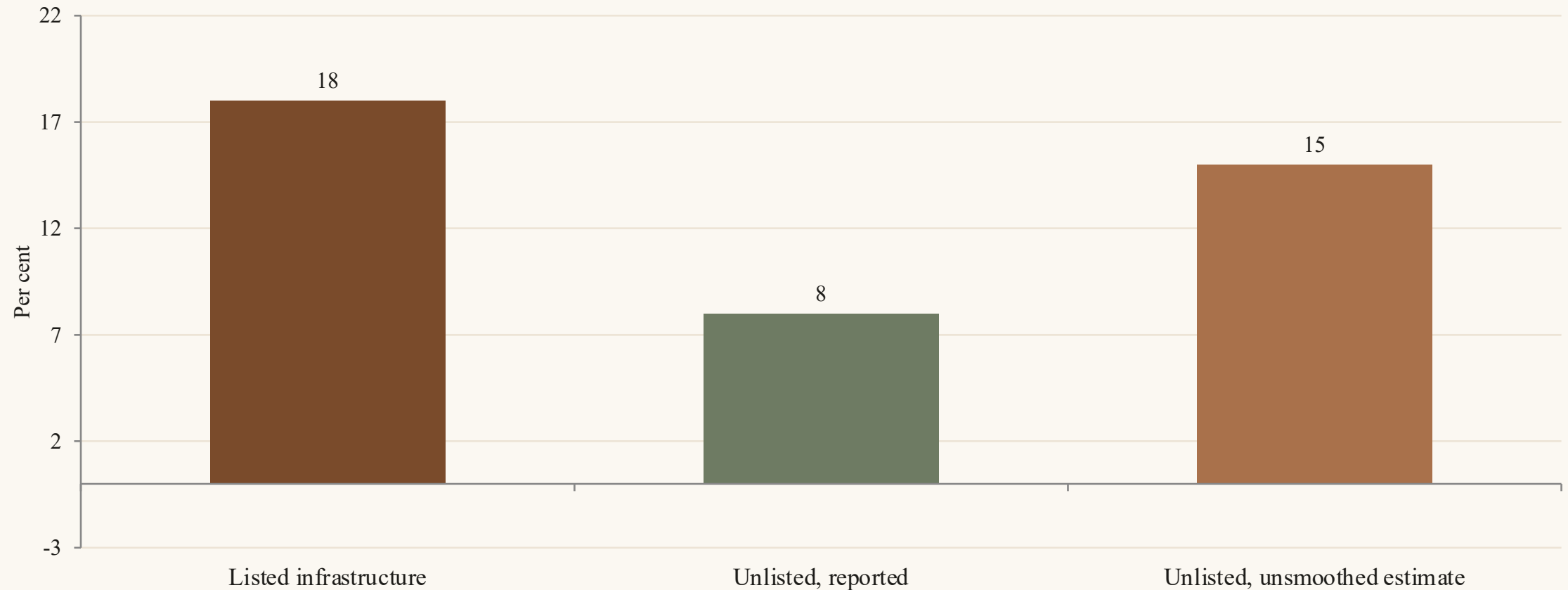
Three problems compound. Unlisted valuations are appraisals rather than transactions, so the series is smoothed and its volatility understated. The reported track record is short and dominated by a single vintage cycle of falling interest rates, which flattered every long-duration asset. And returns are reported by the managers whose fees depend on them, with survivorship and selection working in the obvious direction.

The consequence is not that the asset class is bad. It is that a confident quantitative claim about infrastructure equity returns is not currently supportable, and students should be sceptical of anyone who makes one.

What the appraisal process does to reported risk



Appraisal-based series understate volatility, and unsmoothing removes most of the apparent advantage.



Stylised illustration for teaching only. The figures are not estimates from any series. The purpose of the exhibit is the ordering, which follows from the appraisal process, not the magnitudes.

The illiquidity premium is an assumption more than a measurement

Investors are routinely told that unlisted infrastructure pays a premium for illiquidity. The premium is hard to observe, because the series against which it would be measured is itself constructed from appraisals.

What can be said is narrower and still useful. An investor who does not need liquidity gives up little by forgoing it, so illiquid assets are cheap for that investor relative to one who might have to sell. Whether the market prices that difference as a systematic premium is contested, and in unlisted infrastructure the evidence is thin. Overclaiming in either direction is a mistake.

Internal rates of return do not compare with index returns



A fund reports an internal rate of return. An index reports a time-weighted return. They are different objects and comparing them directly is a category error that appears constantly in practice.

The internal rate of return is money-weighted and sensitive to the timing of draws and distributions, which the manager controls. It can be raised by using a subscription line to defer capital calls without any change in the underlying assets. The standard correction is a public market equivalent, which asks what the same cash flows invested in a listed index would have produced, and it is the only defensible basis for the comparison students will be asked to make in the group project.

The vintage problem

Almost all reported infrastructure fund performance comes from vintages raised in a period of falling interest rates. A long-duration asset bought with cheap debt and sold into a market with cheaper debt still produces a good return whether or not the manager did anything.

Separating manager skill from a duration trade requires either a long enough record to span a rate cycle, which the asset class does not have, or a decomposition of return into yield, operating improvement, leverage effect and multiple expansion. The second is possible and is done rarely, because the answer is often unflattering.

What a benchmark for this asset class would have to do

- Be built from transactions rather than appraisals, or state honestly that it is not.
- Cover a period long enough to include a full interest rate cycle and at least one severe recession.
- Distinguish revenue models, because a regulated network and a merchant airport are not the same asset class in any meaningful sense.
- Be constructed by somebody with no fee dependent on the result.
- Report on a basis comparable with public markets, which means a public market equivalent rather than an internal rate of return.
- Nothing currently available meets all five, which is why week 6 treats benchmarking as an open problem rather than a solved one.

Block C discussion

Seminar questions

1. Andonov and colleagues find a mismatch between the asset and the vehicle. Design a vehicle that fits the asset, and explain why so few of them exist.
2. If the reported track record of unlisted infrastructure is short, smoothed and manager-reported, what would you require before making a ten per cent allocation?
3. Bianchi and colleagues go back to 1927 to find the tail. What does that imply for any asset class marketed on a twenty-year track record?

Block D

Infrastructure debt: credit performance and rating

What forty years of default and recovery data show, and how a forward-looking rating is built.

Indicative length 45 minutes

We know far more about the debt than about the equity

Infrastructure debt has been rated, syndicated and studied for decades, and the loans have been pooled into consortium databases that allow default and recovery to be measured over long periods and across cycles. Infrastructure equity has been reported by managers over a much shorter period.

That asymmetry should shape how students argue. A claim about the credit behaviour of project finance loans can be supported by evidence. A claim about equity returns usually cannot. It is also why the guest hour this week comes from the ratings agencies rather than from a fund manager.

Why project finance debt behaves differently

- The borrower is a single-purpose vehicle with no other business, so there is no contamination from unrelated activities and no diversification either.
- Leverage is high by corporate standards, frequently seventy per cent or more, and is sized against contracted cash flow rather than against assets.
- A network of non-financial contracts allocates construction, supply, offtake and operating risk to counterparties, which substitutes for the covenants a diversified borrower would offer.
- Cash is controlled through an account structure and a payment waterfall, so lenders see and can trap the cash before shareholders reach it.
- Security is over the project and its contracts, and lenders hold step-in rights allowing them to replace a failing operator rather than enforce and sell.
- The consequence is that risk is concentrated in the construction period and declines sharply once the asset is operating, which is the opposite of the corporate pattern.

The consortium data behind the evidence

1983

Start of the period covered by the Moody's study of default and recovery rates on project finance bank loans.

The most cited evidence in this field comes from data pooled by lenders through a consortium, because no single bank has a portfolio large enough to measure default and recovery for a low-frequency event across sectors and regions.

That pooling arrangement is itself the subject of the guest presentation, since Nathaniel Royal directs the Data Alliance at Moody's Analytics.

The headline findings have been stable across editions. Marginal default rates are highest during construction and fall substantially once an asset is operating, and recoveries on defaulted project finance loans are high relative to comparable corporate exposures.

The shape of the risk is what matters

The distinctive feature of project finance credit is not its average default rate but the shape of the hazard over time. Risk is concentrated in construction and commissioning, falls once the asset is operating and revenue is contracted, and rises again only at specific events such as refinancing or the expiry of an offtake agreement.

That shape has consequences everywhere in this course. It explains why greenfield and brownfield are priced so differently, why lenders insist on completion support, why the refinancing point is negotiated so hard, and why an investor able to take construction risk earns a premium that a pension fund cannot access.

An anomaly worth teaching

Hump-shaped

The term structure of credit spreads on project finance loans, against the upward-sloping structure of comparable bonds.

Sorge and Gadanecz compare ex ante credit spreads across a large cross section of international loans and bonds between 1993 and 2001, in both industrialised and developing countries.

For bonds used for purposes other than project financing, the term structure of spreads is approximately linear and increasing in maturity, which is what standard credit theory predicts. For project finance loans it is hump-shaped: spreads rise and then fall as maturity lengthens.

The explanation they offer runs through the sequential resolution of risk. Longer maturities extend into the operating period, when the risks that dominate the early years have already been resolved, so the marginal risk of additional tenor falls.

How a project finance rating is built

Dimension	What the analyst asks	Where a project fails
Financial strength	Do the cover ratios hold under stress, and how much headroom is there	A base case that only works on the sponsor's assumptions
Political and legal environment	Is the jurisdiction stable, and are contracts enforceable	The sovereign ceiling constrains the rating regardless of project quality
Transaction characteristics	Is the technology proven, the construction contract firm, the offtake robust	First-of-a-kind technology, or an offtaker weaker than the project
Strength of sponsors	Can and will the sponsor support the project if needed	A sponsor with no track record or no incentive to stand behind it
Mitigants and security package	What is pledged, and can lenders step in and continue	Security that cannot be enforced in practice in the relevant jurisdiction

The sovereign ceiling, and why it binds

A project cannot usually be rated far above the sovereign in whose jurisdiction it operates, because the state controls the currency, the courts, the tariffs and the ability to move money out. There are structures that pierce the ceiling, generally involving offshore revenue accounts, hard-currency offtakers and political risk insurance, and they are expensive.

The practical consequence is severe and it is the reason block E exists. An institutional investor mandated to hold investment grade paper cannot buy a project in a sub-investment-grade country however good the project is. Capital is not withheld because the project is bad; it is withheld because of where the project is.

The cash waterfall, and why lenders see the money first

Order	Payment	What it protects
1	Operating costs and taxes	The asset keeps running, which is the lenders' real security
2	Senior debt interest and scheduled principal	The primary claim
3	Debt service reserve account top-up	A buffer, usually six to twelve months of service
4	Maintenance and lifecycle reserve	The asset is handed back in specified condition
5	Subordinated and mezzanine debt	The intermediate claim
6	Distributions to shareholders, if lock-up tests are met	What is left, and only if the tests pass

The instruments that make the waterfall bite

- Lock-up tests. Distributions are blocked unless cover ratios exceed a threshold, so a deteriorating project stops paying shareholders before it stops paying lenders.
- Cash sweep. Surplus cash is applied to prepay debt rather than released, accelerating deleveraging when performance is strong.
- Reserve accounts. Debt service and maintenance reserves are funded in advance and can only be drawn for their purpose.
- Step-in rights. Lenders may replace a failing operator and continue the project rather than enforce security and sell an asset nobody wants.
- Together these explain the high recovery rates in the consortium data better than any statement about the quality of infrastructure assets.

Refinancing risk and the mini-perm

Banks will often lend for a shorter period than the asset needs, on the expectation that the loan will be refinanced once construction risk has passed. The structure is called a mini-perm: a facility with a maturity well short of the concession, sometimes with a hard maturity and sometimes with punitive pricing that makes refinancing economically compulsory.

It transfers a real risk to the sponsor. If credit markets are closed at the refinancing date, an otherwise sound project faces default for reasons entirely unrelated to its operations. Several infrastructure defaults have this shape, and it is one of the few points in the life of a project where the risk rises rather than falls.

What we know about emerging market credit, and how we know it

Multilateral and development institutions pool their lending experience through a consortium that records defaults and recoveries on private and sovereign exposures in developing countries over long periods. The resulting evidence is the only large-sample view of emerging market project credit that exists, because individual institutions do not have enough observations.

The recurring finding, and the reason the data have become politically significant, is that realised default and loss experience in these markets is materially better than the sovereign ratings and market pricing would imply. If that is right, the sovereign ceiling is over-binding and the risk perception, rather than the risk, is the constraint. That is a strong claim and it deserves scrutiny rather than acceptance.

Questions to put to the guests

- How much of the difference between project finance and corporate credit performance is the structure, and how much is selection into project finance of assets that were always safer?
- Consortium data solves the small-sample problem and introduces a participation bias. Which banks contribute, and what does the data miss?
- How does a rating committee treat a risk with no historical frequency, such as a physical climate risk over a thirty-year tenor?
- What has changed most in infrastructure rating methodology in the last decade, and what prompted the change?
- For a project in a sub-investment-grade jurisdiction, what actually moves the rating, and how much does each mitigant buy?

Block D discussion

Seminar questions

1. Default risk in project finance is concentrated in construction and falls once the asset operates. Which investors can take that shape, and which cannot, and what does that imply about who funds new assets?
2. The term structure of project finance spreads is hump-shaped. What would have to be true for that to be an arbitrage rather than a correct reflection of risk?
3. The sovereign ceiling binds projects in poor countries regardless of their quality. Is that rational, and what would relax it?

Block E

Why capital does not reach the need

Five barriers, none of which is a shortage of money.

Indicative length 35 minutes

The puzzle, stated precisely

Global savings are abundant. The infrastructure investment need is greatest in low and middle income countries. Capital is mobile. On any simple account, capital should flow towards the need until returns equalise. It does not, and the gap has persisted for decades through periods of very low interest rates when the search for yield was intense.

The explanation is not a shortage of capital and not, mostly, a shortage of returns. It is a set of specific frictions that make an investment either impossible for a regulated institution to hold or unattractive after risk. This block names five.

Five barriers

Currency mismatch

Revenue is in local currency and long-dated finance is available in hard currency. Hedging beyond a few years is unavailable or ruinous in most of the markets that need it.

The sovereign ceiling

A regulated investor holding investment grade paper cannot buy into a sub-investment-grade jurisdiction, whatever the project's own quality.

Political and regulatory risk

Tariffs are set politically, contracts are renegotiated, and expropriation need not be formal to be effective.

Absence of a pipeline

Preparation is expensive, slow and often wasted, so few projects reach the market properly appraised and structured.

Transaction size

A large institution cannot deploy meaningfully in fifty-million-dollar tickets, and the fixed cost of diligence is the same as for a large one.

What they share

None is a shortage of capital, and none is solved by exhortation. Each has a specific instrument that addresses it, which is block F.

Currency is the barrier that is least often solved

A toll road in a middle income country earns local currency. Its lenders want dollars. Someone must bear the difference over twenty-five years, and in most of these markets no one will sell that hedge at a price the project can afford.

The available responses are all partial. Index the tariff to the exchange rate, which transfers the risk to users and is politically fragile. Borrow locally, which requires a domestic capital market with long-dated capacity that mostly does not exist. Use a partial guarantee, which shifts rather than removes the exposure. Or accept a shorter tenor and refinance, which reintroduces refinancing risk at intervals. Delmon's chapter on mobilising local currency financing is the best short treatment of the last of these.

The evidence that structure responds to risk

2012

Hainz and Kleimeier test whether project finance and development bank participation are chosen as responses to political risk.

They find both. Where political risk is higher, a non-recourse project finance structure is more likely to be used rather than a full recourse corporate loan, and a development bank is more likely to be in the syndicate.

The reading is that these are not accidents of practice but instruments. Non-recourse structures limit the sponsor's exposure to a jurisdiction, and a multilateral in the lending group changes the calculus facing a government contemplating a breach.

For a student this is the moment the material stops being descriptive. Contract structure is a response to institutional quality, which is exactly the argument week 2 made from the other direction.

What a pipeline actually requires

- Funded project preparation, which is a public good for the market and is therefore undersupplied by every individual participant.
- Standardised documentation, so that each transaction does not begin from a blank page and legal cost falls with volume.
- A credible forward programme, so that investors and lenders can justify building a local capability.
- A single point of accountability on the public side, so that a developer is not negotiating with four ministries.
- Consistency over time, which is the one element no technical assistance programme can supply.

Ticket size and the aggregation problem

A sovereign fund with two hundred billion dollars under management cannot deploy meaningfully in fifty-million-dollar transactions. The diligence cost is much the same as for a large deal, the governance burden is identical, and a hundred such investments would be needed to move the portfolio at all.

The consequence is that small projects, which is most projects in most countries, are structurally inaccessible to the largest pools of capital. The responses are aggregation vehicles, standardised documentation that lowers the per-transaction cost, and pooled facilities that bundle many small assets into one investable instrument. All three are attempts to manufacture size, and all three have been tried more often than they have worked.

Local capital markets are the durable answer

Every other response to currency mismatch is a workaround. The durable solution is a domestic capital market able to provide long-dated local currency finance, which requires a domestic institutional investor base, an issuance benchmark along the yield curve, and a regulatory regime that permits those investors to hold project exposures.

That is a decades-long institutional project rather than a transaction structure, and it is the reason pension reform and infrastructure finance are more closely connected than either literature admits. Where a country has built a funded pension system, it has also created the natural holder of its own infrastructure.

Block E discussion

Seminar questions

1. Rank the five barriers for a solar project in a lower-middle-income country with a state-owned offtaker. Which would you address first with limited public money?
2. Currency risk over twenty-five years cannot be hedged in most markets that need infrastructure. Whose risk should it be, and what does your answer imply about tariffs?
3. Hainz and Kleimeier show structure responds to political risk. Does that mean project finance is a substitute for institutions, or a symptom of their absence?

Block F

Public support and blended finance

The instruments, what each barrier they answer, and how to tell subsidy from mobilisation.

Indicative length 30 minutes

Instruments, and the barrier each addresses

Instrument	Barrier addressed	Who bears the residual risk
Capital grant or viability gap funding	The project is worthwhile but the tariff cannot cover cost	The state, up front and visibly
Availability payment	Demand risk cannot be borne by the private party at a sensible price	The state, over the contract life
Minimum revenue guarantee	Demand uncertainty, without removing the demand incentive entirely	The state, contingently and often unrecorded
Partial credit or partial risk guarantee	Sovereign ceiling and political risk	The guarantor, usually a multilateral
First-loss or subordinated tranche	The risk-return profile is outside commercial mandates	The concessional provider
Local currency lending or a currency facility	Currency mismatch	The provider of the facility, or the state
Political risk insurance	Expropriation, transfer restriction, breach of contract	The insurer, at a price

Contingent support is the dangerous kind

Grants and availability payments are visible: they appear in a budget and somebody argues about them.

Guarantees and minimum revenue undertakings are not. They cost nothing when granted, appear in no spending total, and become expensive only when the event they cover occurs, which is generally when the state can least afford it.

That asymmetry makes contingent support politically attractive and fiscally hazardous, and it is why the discipline of recording, valuing and reporting contingent liabilities matters more than any other single reform in this area. The instrument is not the problem; the invisibility is.

Does concessional capital actually mobilise

7 to 1

Bank credit crowded in per dollar of multilateral lending over three years, estimated on syndicated loan data for developing countries, 1993 to 2017.

Broccolini and colleagues estimate mobilisation directly. Multilateral participation raises the size of bank inflows, the number of lenders and the average maturity of syndicated loans, and the effects persist for up to three years without being offset by a decline in bond financing.

The estimate matters because the entire blended finance agenda rests on this claim, and until this paper the claim was largely institutional self-reporting.

It is also a ceiling on expectations. Seven to one is a real and useful multiplier. It is not the multiplier implied by official statements that concessional finance will close a fifteen trillion dollar gap.

How to tell mobilisation from subsidy

- Ask what would have happened without the concessional element. This is the additionality question, and the planning students met it in the evaluation of urban policy.
- Ask who captured the concession. If the commercial return is unchanged and the public cost is lower, the concession went to users; if the return rose, it went to investors.
- Ask whether the concession was competed. Concessional capital allocated without competition is priced by negotiation with the beneficiary.
- Ask whether the project would have found capital at a slightly higher tariff. Where it would, the concession bought a lower price rather than an asset.
- Ask what happens next time. A market that learns concessional capital is available for projects of this type will wait for it.

A guarantee is a written option

A minimum revenue guarantee gives the concessionaire the right to be topped up if traffic falls below a threshold. That is a put option written by the state, and it can be valued with the same machinery students met for real options in RE01.

Stating it that way changes the conversation. The state has sold something with a positive value and generally charged nothing for it. It has taken on an exposure that rises when the economy weakens, which is precisely when its own finances deteriorate, so the correlation works against it. And because the option is not recorded, no one is accountable for the price at which it was given away.

Who should bear which risk

- The efficient bearer of a risk is the party best able to influence its likelihood, or to absorb it at least cost if nobody can influence it.
- Construction risk generally belongs with the contractor, who controls it and prices it explicitly.
- Demand risk belongs with the party that can influence demand, which for a toll road is usually neither party, so the question becomes who can absorb it more cheaply. A diversified state usually can.
- Policy and regulatory risk belongs with the state, because the state creates it, and transferring it merely means paying a premium to have it handed back at the moment it materialises.
- Force majeure belongs with neither, and is usually shared explicitly.
- Transferring a risk to a party that cannot manage it does not remove it. It buys an expensive option that will be renegotiated when exercised.

Block F discussion

Seminar questions

1. A government offers a minimum revenue guarantee to attract a toll road concessionaire. Value the guarantee. What is it worth, what should the state have charged for it, and where does it appear in the public accounts?
2. Multilateral lending mobilises about seven dollars per dollar over three years. Is that a good return on public capital, and against what alternative use should it be judged?
3. If concessional capital is available for renewable projects, what happens to the projects that were commercially viable without it?

Block G

The financialisation critique

What happens to an essential service when it becomes a financial asset.

Indicative length 25 minutes

The critique, stated at its strongest

The case against private ownership of essential infrastructure is not that private firms are inefficient. It is that converting a service into a financial asset changes what the asset is for. Ownership becomes geographically remote from the service, returns are extracted through financial engineering as much as through operating improvement, gearing rises because leverage raises equity returns under a regulated revenue stream, and the resulting fragility is borne by users and ultimately by the state, which cannot let the service fail.

Ron Martin's lecture in PGR05 makes this argument in a regional context, connecting financialisation to centralisation and to regional inequality. It is worth taking seriously rather than treating as a complaint.

Two readings of the same facts

The financialisation account

- High gearing is extraction dressed as efficiency, transferring value from users to owners
- Remote ownership severs the relationship between the operator and the place it serves
- The state cannot allow failure, so downside risk is socialised whatever the contract says
- Regulatory settlements are captured over time by the parties with the most at stake and the most resources

The finance account

- Gearing is the efficient response to a stable regulated revenue stream, and the regulator sets the return assuming it
- Ownership location is irrelevant where the regulator sets standards and prices
- The state stands behind the service, not the equity, and equity has been wiped out in real cases
- Capture is a design failure in the regulatory regime rather than a property of private ownership

The housing parallel is close enough to be uncomfortable

The financialisation of housing, which the planning cohort covers in PGR02, describes the same process in a different asset: an essential service acquired at scale by financial owners, managed for total return, with consequences for the people who use it that were not part of the transaction.

Running the two side by side is instructive in both directions. Housing has the better developed critical literature and infrastructure has the better developed regulatory apparatus, and each field would benefit from the other's. For a land economy department this is close to home, which is a reason to teach it rather than an embarrassment.

Remunicipalisation, and what it is evidence of

A number of cities have taken services back into public ownership at the end of a contract or by buying out an operator, most visibly in water. Supporters read this as evidence that private operation failed. Opponents read it as political rather than economic, and note that municipal operators face the same cost pressures with a softer budget constraint.

The honest reading is that both are partly right and that the cases differ. What is not in dispute is that reversal is possible and increasingly attempted, which matters directly to an investor: a concession with twenty years to run in a city where the political consensus has shifted is not the asset that was underwritten.

What the evidence says about ownership and performance

The comparative literature on public against private ownership in utilities does not deliver a clean verdict. Where studies find private operation more efficient, the effect is often attributable to the regulatory regime introduced at the same time rather than to ownership itself. Where they find no difference, the sample frequently mixes very different regulatory settings.

The defensible conclusion is that the quality of the regulatory regime matters more than the identity of the owner, and that a well-regulated private operator and a well-governed public one outperform a badly governed version of either. That is less satisfying than a verdict and it is what the evidence supports.

Block G discussion

Seminar questions

1. A regulated water company is highly geared and pays substantial dividends while missing environmental targets. Is that extraction, or is it the regime working as designed with the targets set too loosely?
2. Does it matter where the owner of a national network is located, given that a regulator sets prices and standards?
3. If the state cannot allow an essential service to fail, has any risk genuinely been transferred to the private sector? What would have to be true for the answer to be yes?

Five things to look for in any infrastructure investment proposal

The revenue model

Regulated, contracted or merchant.
Everything about the risk follows from this and it is often stated less clearly than the return.

The contract expiry

When does the offtake, concession or subsidy end, and what is the asset worth the day after. Terminal value assumptions are where optimism hides.

The exit assumption

Who buys this in year ten, at what multiple, and what has to be true about the market then for the return to be realised.

The indexation

Which index, applied to what share of revenue, with what lag, and until when. Partial indexation is routinely described as inflation protection.

The counterparty

Who actually pays, and what is their credit. A contracted revenue stream is only as good as the entity contracted to pay it.

Where to find them

Not in the summary. These are in the schedules, and the discipline of reading them is most of what separates a good analyst from a fast one.

A note on data sources students will actually use

For listed exposure, market prices and company reporting are available and reliable. For unlisted funds, the principal sources are commercial databases assembled from voluntary manager reporting, with the selection problems that implies. For infrastructure debt, the ratings agencies publish default and recovery studies and the consortium databases sit behind them. For emerging markets, the multilateral consortium data is the only large-sample source.

None is complete and each has a known bias. Naming the bias when citing the source is the difference between a defensible claim and an assertion, and it is what the individual coursework will be marked on.

Week 3 synthesis

Seminar questions

1. You advise a mid-sized pension scheme with a twenty-five year liability profile and no internal infrastructure capability. Recommend a route into the asset class, quantify what your recommendation costs relative to direct ownership, and defend it against the objection that the fee load removes the premium.
2. A government asks how to attract private capital into a water sector project in a lower-middle-income country. Using the five barriers and the instrument table, set out what you would do first and what you would deliberately not do.
3. Is the case for private investment in infrastructure primarily about capital, capability or discipline? Which of the three does the evidence in this week actually support?

Block H

Reserve material

Slides held back from the main sequence.

Indicative length As required

Candidate cases for Week 3

Case	What it illustrates	Material
A large listed infrastructure fund	Listed against unlisted, and the discount to net asset value	Annual reports and market prices are public
A Canadian pension plan's direct programme	The make-or-buy decision at scale	Extensive public reporting
An emerging market solar programme with a guarantee	Blended finance and the instrument-to-barrier match	Multilateral project documents are public
A regulated water company's capital structure	Financialisation, gearing and the regulatory response	Ofwat and company disclosures
An offshore wind transmission tender	A regulated asset created by competition	Ofgem tender evaluations

An alternative running order

Where the guests are presenting for a full hour rather than two short slots, compress blocks A and B into a single half-hour framing and run block D immediately before the guests, so the framework is fresh. Blocks E and F then follow the presentations and benefit from the credit material.

Where the cohort is drawn mainly from Planning and Environmental Policy, invert the order: open with block E on why capital does not reach the need, use that to motivate the investor's problem in block A, and treat block D as background rather than as a teaching block.

Sources drawn on for this deck

- Weber, B., Staub-Bisang, M. and Alfen, H. W. (2016), Infrastructure as an Asset Class, chapter 2.
- Gatti, S. (2024), Project Finance in Theory and Practice, chapters 1, 6 and 8.
- Abor, J. Y. and others (eds) (2025), The Routledge Handbook of Infrastructure Finance, chapters 12, 14, 15 and 16.
- Delmon, J., Private Sector Investment in Infrastructure, chapters 10 and 11.
- Flyvbjerg, B. (ed.) (2017), The Oxford Handbook of Megaproject Management, chapter 16.
- Journal articles as cited on each slide. Full references in the module reading list.

Where Week 3 leaves us

An allocation decision is a liability-matching decision, and the route into the asset determines the return as much as the asset does.

We know a great deal about the credit performance of infrastructure debt and much less about equity returns than the marketing implies.

Capital does not reach the greatest need because of currency, the sovereign ceiling, political risk, pipeline and ticket size, none of which is a shortage of money.

Whether private ownership of an essential service is efficiency or extraction is a real argument, and both accounts are internally coherent.